Smart Acoustic Sensor

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***Abstract*—To automatically detect dangerous situations in public areas, systems based on visual detection are the most commonly used. In certain situations, to detect a source of danger, an automatic acoustic detection system may be preferable. This paper presents a smart acoustic sensor system, which automatically detects acoustic pulse events, such as gunshots, in public places, and localizes the position of the acoustic source. Tested firearms are 9 mm short gun, 6.35 mm short gun, .22 short gun with various subsonic and supersonic ammunition. Various sounds like door slams, human screams, etc. represent false alarms.**

Keywords*— *acoustic impulse signal, acoustic sensor, gunshot detection, signal processing

I. Introduction

In these days, acoustic impulse signal detection techniques are required not only in military applications but also in surveillance public areas such as schools, campuses, shopping centers or hospitals. Detection, localization, and classification of acoustic impulse signal (gunshots) is of interest in the above mentioned areas. There are several commercial or experimental acoustical detection and localization systems designed for gunshot occurrence available on the market [1] - [5]. These systems are intended to detect acoustic impulse signal – gunshot and to localize the direction of the source of the signal. Systems that are more sophisticated can even classify and identify the particular firearm type. On the other hand, a drawback of these systems is high purchase and operational cost. For example, in [6] uses feature extraction based on mutual information in both time and frequency domain, in [7] uses a wavelet transformation and advanced classification using SVM or k-nearest neighbors classifiers. All these methods demand great computational means and therefore increasing the cost of the detector.

To detect and classify a gunshot, a better understanding of its complexity physic have to be understood first. In [8] and [9] a comprehensive explanation is presented. Gunshot pattern is produced by two phenomena: The muzzle blast, and, if the bullet travels at supersonic speed, shock wave. The muzzle blast is caused by the rapidly expanding gases from the confined explosive charge that is used to propel the bullet out of the gun barrel and propagates through the air at the speed of sound. The shock wave propagates away from the bullet's path and expands as a cone behind the bullet, with the wave front propagating outward at the speed of sound [8]. The shape of a gunshot acoustic impulse signal depends on many factors such as the caliber of both bullet and barrel, the length of the latter, or even the chemical properties of the propellant. The acoustic impulse signal also depends on multiple sets of factors such as a temperature of the air, air humidity, wind speed, environment (e.g., foliage density, urban area) and soil characteristics [10]. Above that, the measured acoustic impulse signal also contains information about the surroundings. Figs. 1 and 2 shows a typical representation of a subsonic and supersonic gunshot signals.

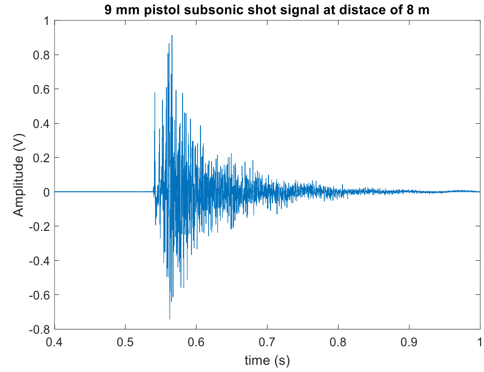

Fig. 1. Signal corresponding to 9 mm pistol subsonic shoot at a distance of 8 m

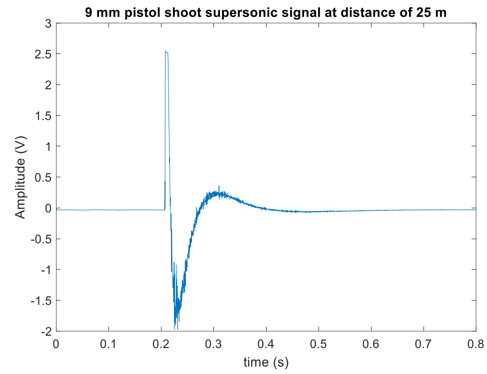

Fig. 2. Signal corresponding to 9 mm pistol shoot supersonic at a distance of 25 m

In Fig. 2, due to the relatively low speed of the supersonic bullet (Mach number $M = 1.09$), the shock arrival is not visible thanks to its proximity to the muzzle blast part of the signal.

Considering all these criteria, to successfully identify and detect a gunshot impulse signal, advanced signal processing including adaptive filtering and complex data classification have to be deployed. In this article, a Smart Acoustic Sensor (SAS) able to detect an incoming gunshot in a noisy environment is presented. The system determines the azimuth and elevation of the incoming detected gunshot. The system consists of four microphones – three for the azimuth placed in the base of the tetrahedron and one for elevation, placed on the top, analog signal pre-processing unit and digital signal processing and data analysis unit. Compared to other existing available solutions (e.g. [3] or [5]), the proposed system has a unique modular structure which means its deployment is quite flexible (e.g., the sensor part can be located on the building or car roof while the processing unit can be installed in a distant protected place). It is also quite cost-effective as it is composed mostly of inexpensive commercially easily available components. Even though, its parameters are fully comparable with professional systems deploying proprietary dedicated HW components. In the future, these systems can be used as a standalone units placed in schools, campuses, shopping centers or public areas, in general, to detect and localize gunshot events. The units can also work as system element, delivering data into the shared central server. Such topology allows for additional analyses, i.e., triangulation localization of the event position using time-stamped data from multiple units receiving the acoustic signal related to the shooting.

The paper is organized as follows. In Section II, the SAS system and detection algorithm with the modified median filter are introduced. In Section III, experimental and real data together with the detection algorithm results are presented. The conclusion and future work are presented in section IV.

II. METHODS

There are many methods for dealing with the detection of gunshots [11] – [17]. The presented SAS use modified median filter method described below. The acoustic impulse signals are recorded using microphones placed in the tetrahedron (Fig. 3). In this case, the distance *x* between microphones in the tetrahedron base is 54.0 cm, while the height *h* of the tetrahedron is 44.1 cm.

Fig. 3. Acoustic sensors placed in tetrahedron

The acoustic signals are amplified by the four-channel amplifier (RME QuadMic II) and digitized using four-channel 16-bit digitizer (NI9222 placed in NI cDAQ 9171 chassis). The digitized data (all four channels) are processed in parallel threads of the detection algorithm. Each acoustic signal is firstly squared, thus converted to energy signal. To filter the various noisy environment background, the signal has to be filtered. Advanced filtering method such as adaptive filtering method is one of the options. Therefore, the acoustic energy signal is then filtered by the modified median filter. The acoustic energy signal is divided into the predefined odd number of blocks of data – windows. The middle window is the so-called actual data window, where the detection is performed. Each window represents 50 samples, i.e., 1 ms of recorded data long. The data in the window before the window of actual represents the acoustic signal recorder after the actual data and vice versa, data recorded before are stored in windows which follows actual data (Fig. 4). A number of windows changes the performance of the median filter. The optimal number of windows (12 + 1 with actual data) has been set by experimental measurements. These windows served as an input of the modified median filter.

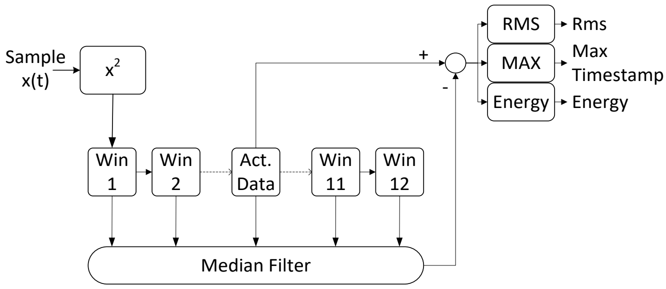

Fig. 4. Block diagram of the median filter

The filter output is the noise background of the acoustic signal energy which is subtracted from the middle window with actual data. The middle window with filtered actual data is then tested for a gunshot event. To distinguish true acoustic event from false alarms, the actual window is searched for a maximal peak which meets the following criteria:

- Maximal value of the signal has to be greater than *Sensitivity Threshold Level* (STL) parameter
- Maximal value of the signal has to exceed RMS of the signal multiple times (*RMS Threshold Level*)

These parameters adjust detector sensitivity, and their values depend on the environment. Both parameters are set adaptively to suppress false detection due to the noise of the environment. When the acoustic event meets both these criteria, moreover the algorithm compares the energy of the potential gunshot signal with the Energy Threshold Level (ETL). In case the energy exceeds the ETL algorithm compare the overall energy of the window before and after the actual data window. If the energy of the signal belonging to the window after the detected maximum is greater than the energy of the signal belonging to the window before the window where the acoustic event is detected, the gunshot is indicated.

The next step of the algorithm analyses the maximum acoustic signal delay between two microphones. To avoid false detections caused for example by multiple consequent impulse sounds from different directions, the maximum time delay of an acoustic signal between microphones is computed. If the detected gunshot delay between microphones is greater than the theoretically calculated delay, the shot is not indicated. The output of the detection algorithm is four timestamps (one for each microphone/thread). The algorithm can be expressed by following pseudocode (Tab I).

TABLE I. THE ALGORITHM PSEUDOCODE

```
If (Max > Sensitivity Threshold Level) &&
        (Max > (Rms * RMS Threshold Level))
        {
               If (energy (Actual Data Window) > Energy Threshold
               Level)
                      {
                             If (energy (Actual Data Window + 1) > energy
                             (Actual Data Window - 1))
                                    {
                                      Gunshot Detected (t1, t2, t3, t4)
        }             }             }
```

III. RESULTS

Tested firearms has been 9 mm short gun, 6.35 mm short gun, .22 short gun, .22 rifle gun. A various subsonic and supersonic (up to M = 1.1) ammunition has been used with the 9 mm short gun. As false alarms, a various door slams, handclaps, human screams, dog barks, etc. has been used. 41 different signals corresponding to gunshots and 80 signals corresponding to false alarms has been recorded.

Firstly, the experimental data has been taken in the anechoic acoustic chamber and recorded by proposed SAS (see chapter II). The distance between the shooter and the acoustic detector has been 8 m. The sampling frequency has been set to 250 kSa/s. The example of recorded and filtered subsonic gunshot signal, taken in the acoustic chamber from all four microphones is in Fig. 5. The presented signal is converted from voltage signal to the energy signal by powering to two of the original signal.

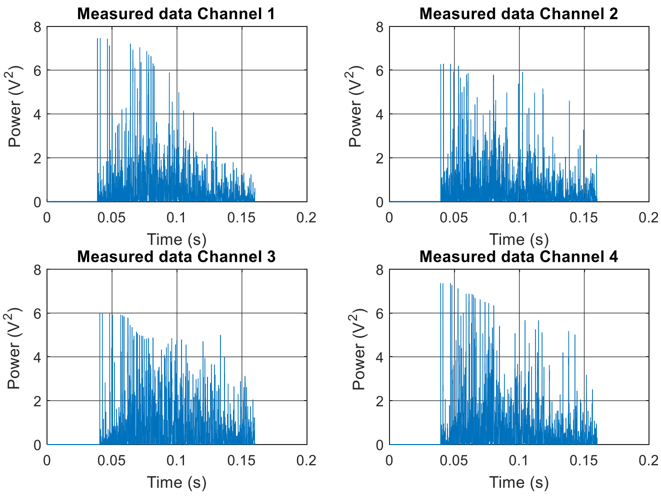

Fig. 5. Signal corresponding to subsonic gunshot recorded by the Acoustic Smart Sensor

The microphones placed in the base of the tetrahedron are numbered as follows. The microphone (resp. corresponding thread) which detect the gunshot as first is numbered as a first (channel 1), the microphone, which detects the gunshot as second, is numbered as second, etc. While the microphone on the top of the tetrahedron used for the elevation calculation is always numbered as a fourth. These experimental data have been used to set an optimal number of windows for modified median filter and presets of all parameters for detection algorithm. The real data has been taken at shooting range at different distances from 10 m to 25 m, using the configured hardware setup. As a background noise, a diesel engine in different distances together with loud speech has been used. The example of recorded subsonic .22 caliber short gun gunshot signal, taken in a shooting range noised by the diesel engine is in Fig. 6 (see the difference between noisy background in Fig. 6 and quiet background in Fig. 1). The sampling frequency of the recorded signal in real conditions has been set to 50 kSa/s. Only one channel is presented in Fig. 6.

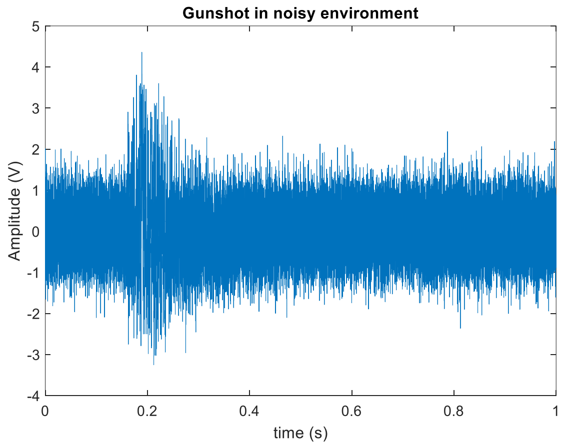

Fig. 6. Signal corresponding to gunshot of .22 caliber short gun in noisy environment (SNR = 5dB)

The example of recorded signal corresponding to supersonic 9 mm caliber short gun gunshot is Fig. 7. Again, the signal is filtered and converted to the energy signal.

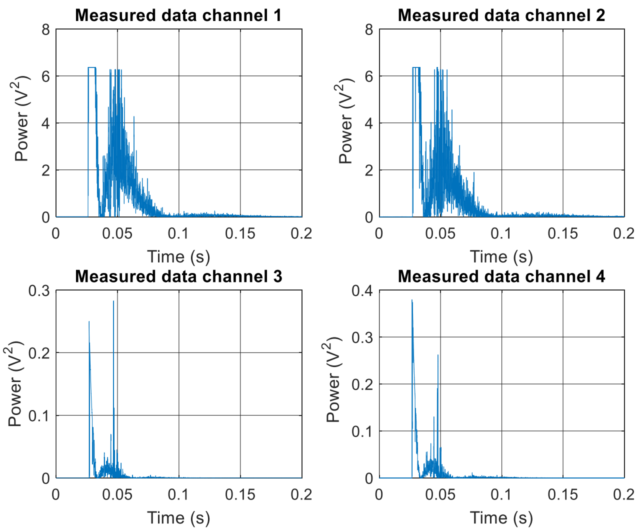

Fig. 7. Signal corresponding to supersonic gunshot recorded by the Acoustic Smart Sensor

If the algorithm (all four threads), presented in chapter II, detect a gunshot, Smart Acoustic System calculates the azimuth and the elevation of the incoming shot from the four timestamps. For azimuth and elevation computation (using the four timestamps) the following parameters are defined:

- τ_1 is the time difference between the first occurrence and the second occurrence of gunshot detection timestamp
- τ_2 is the time difference between the first occurrence and the third occurrence of gunshot detection timestamp
- τ_3 is the time difference between the first occurrence of gunshot and gunshot detection timestamp corresponds to the signal from the elevation microphone
- τ_M is the computed maximal time delay between two microphones

The azimuth is calculated by formulas:

$$\alpha_{1,2} = \arccos\left(\frac{\tau_1}{\tau_M}\right) \tag{1}$$

$$\alpha_{3,4} = \arccos\left(\frac{\tau_2}{\tau_M}\right) + 60° \tag{2}$$

Both (1) and (2) have multiple solutions due to harmonic function periodicity. However, the right solution is the one that is common both for (1) and (2):

$$\alpha_{1 \text{ or } 2} = \alpha_{3 \text{ or } 4}. \tag{3}$$

The elevation of incoming gunshot is calculated by (4).

$$\beta = arctg\left(\frac{v_\alpha}{h}\right) - \arcsin\left(\frac{v_s \tau_3}{\sqrt{h^2 + v_\alpha^2}}\right) \tag{4}$$

,where v_s is the speed of sound and v_α is correction factor for the height of the tetrahedron given by formula:

$$v_\alpha = \frac{\sqrt{3}}{4} x + \frac{1}{6} x \frac{\sqrt{3}}{2} \cos(3(\alpha - 30°)) \tag{5}$$

The gunshot detection is presented by SAS to the user with the graphical user interface. The Main window contains the following elements: the computed azimuth and elevation of the gunshot detection, circular section indicates azimuth/elevation with scattering ±5° (for better visual orientation) and labels with display the temperature and the relative humidity, along with the calculated sound speed. GUI of the SAS is shown in Fig. 8.

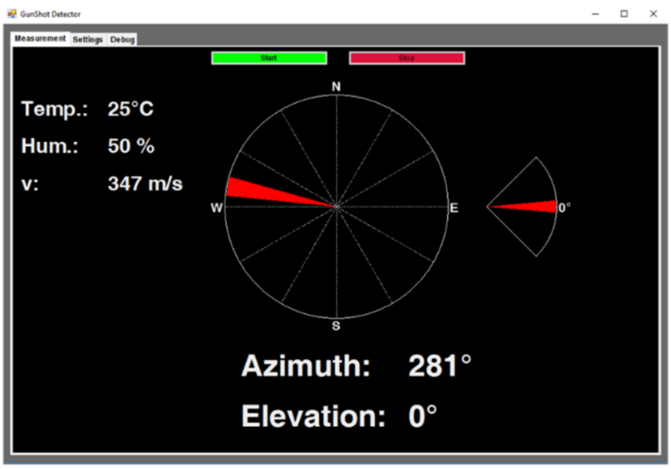

Fig. 8. GUI of the Application

Total uncertainty of the system (for the correct input parameters) is ±1.8° for azimuth and ±3.8° for elevation. The total uncertainty of the system depends on the frequency of the oscillator (10^{-6}), the sampling frequency, the uncertainty of the distance between the microphones in tetrahedron (±3 mm) and the uncertainty of the speed of sound.

In total 41 signals corresponds to subsonic and supersonic gunshots (see examples of the corresponding signals presented on Fig. 5 and Fig. 7) and 80 signals represent false alarms (various doors slams, handclaps, scream, dog bark ...) have been tested on presented SAS. The example of one of the false alarms – a door slam is presented in Fig. 9.

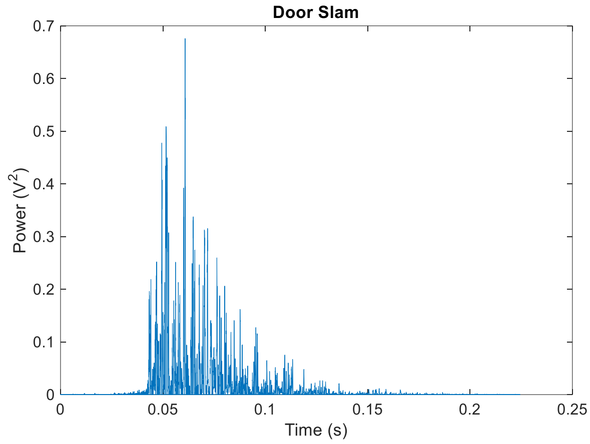

Fig. 9. Signal corresponding to door slam recorded by the Acoustic Smart Sensor

The signal corresponding to door slam is filtered and converted to the energy signal using the same algorithm as previously presented gunshot signals.

The test results for all signals in the form of True Alarms Occurrences (TAO) and False Alarms Occurrences (FAO) are shown in Tab. II. 100% of gunshots has been correctly detected, on the other hand, 12% of signals corresponding to a door slams has been evaluated as a false alarm.

TABLE II. THE RESULTS OF THE DETECTION

	gunshot	*door slam*	*handclap*	*scream*	*dog bark*	*others*
TAO	41/41	/	/	/	/	/
FAO	/	3/25	0/15	0/15	0/15	0/10

From the experimental and real data measurements the Smart Acoustic Sensor, operating on the principle of the modified median filter, can successfully detect an acoustic event and can recognize a gunshot signal up to Signal to Noise ratio SNR ≥ 5dB. If the SNR is lower than 5 dB, the detection success rate drops dramatically.

IV. CONCLUSION

In this paper, the SAS for a detection of gunshots has been presented. It is based on a modified median filter. The system can detect and localize the azimuth and elevation of the incoming gunshot to SNR ≥ 5dB. Experimental measurements, taken in the anechoic acoustic chamber were conducted to set parameters of the proposed algorithm to increase performance and reduce false alarms. The real measurements were taken at a shooting range using different firearms, and various subsonic and supersonic ammunition showed the detector could detect the tested gunshots in 100% occurrence.

In the future, currently, limited datasets will be extended by new measurements, which are now ongoing. Moreover, the recognition and classification of the caliber of the firearm will be evaluated and implemented to the SAS.

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